PresenterX — System Architecture & Workflow Overview

This document provides a technical blueprint of the generation graph, visual engine algorithms, and layout processing nodes of PresenterX.

1. LangGraph Orchestration Pipeline

PresenterX processes prompts and coordinates visual rendering stages asynchronously via a multi-agent system state machine:

- **Input Node**: Initialized Presentation OS state context variables and tracks execution metrics in database logs.
- **Research Node**: Scrapes websites or YouTube URLs, then triggers Tavily RAG engines for targeted search context summaries.
- **Parallel Generation Node**: Concurrently plans visual layout categories, color themes, keywords, and structural outline cards.
- **Slide Validator Node**: Calculates semantic completeness scores, automatically triggering remedies if parameters score below 90.
- **Layout Engine Node**: Leverages specialized pattern generators (feature grids, timeline nodes, TAM charts) to map contents.
- **Design Critic Node**: Runs visual compliance checks to correct low contrast, adjust spatial overlap, and scale font sizes.
- **PPTX Compiler & Export Node**: Constructs finalized slides, dynamically retrieves stock images, and writes to PPTX.

2. Slide Pattern Engine & Visual Selection

PresenterX maps copywriter texts directly into visual components rather than plain bullet-point layouts:

- **Hero Layout (Cover)**: Full-bleed background assets with bold centered headings and primary subtext descriptions.
- **Feature Grid**: Divides parallel concepts into custom-spaced containers using 2px strokes and drop shadows.
- **Comparison Matrix**: Displays features across side-by-side matrices with highlight markers for visual emphasis.
- **Timeline & Roadmap**: Processes chronological dates or deliverables across swimlanes or horizontal processes paths.
- **Market TAM**: Models target addressable sizes visually using Concentric TAM/SAM/SOM circles and progress CAGR growth blocks.
- **Architecture Layer**: Connects related services and APIs together into vertical node layer flows (Frontend to Database).

3. Design Compliance & WCAG Guidelines

A post-processing layout layer ensures every generated file remains readable and structured:

- **WCAG 2.1 AAA Contrast checks**: Measures relative luminance using target R, G, B color vectors. If color contrast relative to background falls below 4.5:1, text colors are forced to high-contrast fallback dark blue or white.
- **Strokes & Shadows**: Always applies clear 2px border strokes and soft drop shadows to layout containers for legibility.
- **Reading Scalability**: Enforces minimum font size boundaries (Title headers 36px, Card labels 18px, body text 14px).
- **Viewport Occupancy**: Validates layout density. Rejects slides with viewport coverage below 55%, triggering a high-density recalculation.